

RAGAVI M

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Professional Experience

A highly self-motivated professional with over **9** years of experience in **Embedded** Systems Development specializing in the **Set-Top Box (STB)** middleware domain. Adept in working on both **Android** and **Linux** platforms.

- **Senior Software Engineer** | October 2019 to Present | **L&T Technology Services - Chennai, IN.**
- **Associate Software Engineer** | May 2016 to September 2019 | **Sure Soft Systems Pvt Ltd - Pondicherry, IN.**

Technical Proficiency

- **Programming Languages:** Proficient in C and Embedded C; working knowledge of C++ fundamentals(OOP's) with conceptual understanding up to C++14, including, STL and design patterns.
- **Operating System:** Linux, Android, Windows.
- **Linux Internals:** Device drivers, File systems, Memory management, Cross-compilation, Kernel debugging.
- **Communication Protocols:** I2C, UART
- **STB Standard:** DVB & ARIB Standard.
- **Debugging Tools:** GDB, Valgrind
- **Kernel Debugging Tools:** KGDB, ftrace, strace.
- **Build Automation:** Makefile, CMake.
- **Delivery Systems:** DVB-S/S2, DVB-T/T2 and DVB-C.
- **Source Management:** GIT, SVN.
- **Bug Tracking:** JIRA

Experience Summary

- Proficient knowledge & experience in C and C++ language(basics) and Linux Internals.
- Good working experience in the field of STB full stack middleware developer.
- Having work experience in porting the device driver (BCM43569 Wi-Fi dongle in. set top box).
- Customized Linux kernel based on specific product requirements
- Experience in developing and Integrating Linux device drivers (char device driver).

- Skilled in kernel debugging (like kernel panic issues) and performance optimization.
- Knowledge in Kernel Internals and experience in Boot time optimization.
- Basic knowledge in Boot process and Bootloader.
- Knowledge in kernel debugging utilities(dmesg, syslog).
- Extensive knowledge of DVB Standards and ARIB standards.
- Good working experience in the DVB-S/S2, DVB-T/T2 and DVB-C delivery systems.
- Having good work experience in Own Tool generation for create/modify transport stream details based on the given input.
- Proficient in working with various PC-based tools.
- Excellent problem-solving skills and debugging skills.
- Coordinated the team of 5 professionals for their bug solving and for code review
- Effectively managed customer communication (in weekly status meetings) and coordination of the project development activities (both internal and external).

Android Project Profile

- 1. Android to RDK Migration (POC)** - Upgrading Android OS to RDK through OTA process. - Handled the Bluetooth component in this part.
Responsibilities:

- Exploring the knowledge on Bluedroid and Bluez stack.
- Acquired knowledge of Bluetooth connectivity operation (scan, pair and unpair) on both bluedroid and bluez stack.

- 2. Canal Digital (BCM7251S)] DVBS/S2x4 Tuner + Android TV + PVR + Conax CAS**
Responsibilities:

- Solved the EPG schedule event loading issue.
- Designed and developed character device driver (DLPC3433 display controller) for Vantiva video device - POC.

- 3. DISHTV Super Box (Hi3798CV200)] DVBS/S2x4 Tuner + DVBT Tuner + Android TV + PVR + Google Certified**
Responsibilities:

- Collected the EPG event information and stored in SQL Database in TIF(TV Input Service) compatible formats.

Linux Project Profile - ARIB Based

1. JCTA - Japan Cable - Linux based STB project which provides DVB-C, DVB-T and DVB-S broadcast services reception and AV rendering.

Responsibilities:

- Exploring the knowledge on ARIB standard which is frequently used by Japan country.
- Solved the tuning and recording related issues reported by Test group and the customer claim issues.
- Solved the memory leak issues in Middle-ware level.
- Solved the issues in browser module and understood the HTML5, TTML5 application functionalities.
- Solved the customer claim issues like STB freeze, hang and HDMI related issues.
- Porting the new Wi-Fi dongle (BCM43569) to this JCTA project. And integrated the same into this platform.
- Prepared the Weekly Status Report and attending and reporting the weekly discussion with client.
- Supported our team members for code review and bug solving.

Linux Project Profile - DVB Based

Linux STB: The project involves implementing and enhancing various features for Linux-based Set-Top Boxes (STBs), including tuning functionality, service management, LCN assignment, HBBTV support, and more, based on broadcast and network data.

Project list:

1. CHANGHONG Hi3796MV200
2. RURUTEK 7C123 with NAGRA CAS
3. DTM Hi3798MV200 Freeview Plus
4. TELE MSD7S235 SUN Direct DVB-S
5. DISHTV DVB-S and DVB-T
6. WILLET- DVBC

Responsibilities:

- **Transponder Tuning & Service Collection:** Implemented home transponder tuning and service collection based on NIT and BAT tables, ensuring accurate channel and frequency alignment.
- **LCN Assignment:** Designed and managed LCN assignment workflows for both DVB-T and DVB-S, including conflict resolution and simulcast service management for 1000+ LCNs.
- **EPG Data Processing:** Supported EPG data collection from multiple table formats, including backward and forward event guides (past 7 days and future 7 days). Implemented optimization techniques for schedule sorting and event search.

- **OTA Download & Bouquet Management:** Enabled parsing and signalling of OTA details and optimized bouquet table updates for dynamic network changes.
- **Event Booking & Recording Issues:** Resolved complex event booking challenges, including alternate event booking, overlaps, and partial/failed recordings.
- **Customer-Specific Customization:** Communicated with clients to understand requirements, implemented tailored solutions, and resolved end-customer issues for multiple operators.
- **Freeview Service Integration:** Integrated country-specific features for Freeview, including service collection and EPG enhancements.
- **Dynamic Descriptor Support:** Supported bootup LCN descriptor updates by identifying and managing dynamic changes.

Academic Qualification

2024 - MCA from Alagappa University at Karaikudi – Still pursuing the degree (1st year completed)

2016 - B.Sc (Computer Science) from Ayya Nadar Janaki Ammal College, Sivakasi with an aggregate of 80%

Declaration

I hereby declare that the information and the facts stated above are true and correct to the best of my knowledge.

Place: Chennai

Date:

M.Ragavi